

The Effect of Medicaid Expansion on Ischaemic Heart Disease Mortality

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I. Abstract

Medicaid Expansion gave millions of low-income adults across the United States access to health care, yet its effect remains debated. This paper aims to estimate the effect Expansion had on mortality from heart disease for middle-aged adults. By using economic and policy controls, we compare states that adopted Expansion to those that have not. While the results did not produce statistically significant effects that were distinct from zero, we were able to rule out large effects. As many positive effects from Expansion will be undone by Trump's One Big Beautiful Bill, it's especially important to fully understand the consequences of losing health insurance.

II. Introduction

Before diving into our results, it's important to lay the groundwork for why chronic diseases and healthcare are correlated. Chronic diseases are those that last over a year and require ongoing care. This means multiple doctor visits per year and taking medications frequently. Some common chronic diseases are heart disease, high blood pressure, and dementia. They are also very widespread in the United States, as three in four adults enrolled in Medicaid have one or more chronic conditions (Kaiser Family Foundation, KFF 2025). This now brings us to Medicaid, which is a joint federal and state insurance program that gives healthcare to low-income individuals and households. Health insurance allows its users to receive health care and medications for free or at a significantly lower price. As expected, those without health

insurance are more willing to skip receiving care or regular doctor check-ups as a result of high costs. Indeed, data shows that three in four uninsured adults have skipped or postponed healthcare in the past 12 months due to cost (KFF 2025). In light of this, chronic diseases are especially “easy” to overlook because they happen gradually and are easy to put off for longer periods of time.

Next, we consider Ischaemic Heart Disease and Medicaid Expansion. Ischaemic Heart Disease is a chronic condition that is the leading cause of death in the US (CDC 2026). It’s caused by plaque building up in the arteries and mostly affects adults over 65 (Memorial Hermann 2025). That said, over the past 15 years, there’s been a noticeable increase in the number of people affected who are less than 50. Secondly, we consider Medicaid Expansion. Medicaid Expansion was a federal policy to expand the number of people who could receive Medicaid by expanding eligibility to adults with incomes up to 138% of the poverty line (KFF 2025). As of June 2024, this policy has given Medicaid to 20 million additional people. This number is not insignificant, as it represents nearly a quarter of all Medicaid enrollees. Importantly, not all states adopted Medicare Expansion. As of today, 40 states and the District of Columbia have adopted Expansion. Given Trump’s recent One Big Beautiful Bill, which will cut Medicaid access to many, now is an important time to consider the effects Medicaid Expansion has had so far.

This paper asks, did Medicaid Expansion reduce mortality from Ischaemic Heart Disease for adults aged 55-64? This group of adults is significant because they don’t yet qualify for Medicare but still have a meaningful risk for heart disease. Medicare is the US joint federal and state program that gives healthcare to those aged 65 or older. As discussed, health care significantly increases the chance that people receive care. This care would take the form of

taking medication or a doctor's visit, sharing the risks of smoking and a poor diet for a particular individual.

We answer this question with a difference-in-difference of state-year panel data from 1999 to 2019. Essentially, we compare mortality trends from Ischaemic Heart Disease in states that never adopted Medicaid expansion to those that did. For each state and year in the given range, we consider the crude rate per 100,000, which is the number of people who died from Heart Disease, given a 100,000 people in the population. We collected data from five main databases. Firstly, we found crude rates from the CDC Wonder Cause of Death. Secondly, we found the year each state adopted or never adopted Expansion from the Kaiser Family Foundation. The final three sources were used to collect data on our controls, which were unemployment rates, poverty rates, and cigarette tax rate for each state.

Ultimately, when testing for the specific effect of Medicaid Expansion on crude mortality from Ischaemic Heart Disease, we failed to find a statistically significant effect that was different from zero. Even with a plethora of subgroups, such as only men or only those aged 55-64, and different combinations of controls, such as only state and time fixed effects or excluding the cigarette tax, the effect was largely insignificant. That said, a null result still provides insight, and there were insightful dynamics between regressions that we will explore in the results section.

Current papers have broadly found positive health outcomes from Expansion. A 2019 JAMA paper finds that counties with Expansion have 4.3 fewer deaths per 100,000 people from cardiovascular causes (Khatana 2019). It also found significantly lower rates of cardiovascular mortality for middle-aged adults in Expansion states. More broadly, a 2019 paper from IZA found decreases in all-cause mortality of 3.6% (Borgschulte 2019). This paper differs from the previous two papers in two key ways. The First difference is in scope of disease. While the

JAMA paper considers all cardiovascular causes and the IZA paper considers all-cause mortality, this paper focused on Ischaemic Heart Disease. Our assumption is that this disease should be significantly affected by health insurance. The second difference is in specificity of analysis. While the previous two papers consider the county level, this paper tests whether results are visible at the state level. Thus, the contribution of this paper is to provide evidence of a possible effect from Expansion for a specific cardiovascular disease at the state level by investigating the effect on mortality from a specific, chronic heart condition.

The remainder of the paper proceeds as follows. Section III. Data describes the data sources. Section IV. Empirical Methods explains the difference-in-difference and regression equation. Section V. Estimation describes the results. Section VI. Conclusion recaps the findings and provides limitations of the paper. Finally, Section VIII. Appendix includes all graphs, regression tables, and figures.

III. Data

This Data section will describe the five data sources used in this paper and the manipulations made to the data; however, before examining specific sources, it is first important to understand what type of structure we want to get our data into. Since our end goal is to run a difference in difference, we need to get our data in a state-year panel format. This simply means each row of our data will be a unique state and year combination. By state, we consider all US states and D.C., and by years, we consider the years 1999-2019. One example of a state-year combination would be Massachusetts in 2004. Continuing this example, each state-year combination would have additional information, such as whether Medicaid Expansion had happened or the crude mortality rate from heart disease. These additional pieces of data can be

thought of as columns, and one data point in the column will be related to the row for Massachusetts in 2004. In total, since we have 51 states, including DC, and 21 years to consider, we need data for $51 \times 21 = 1,071$ rows of data.

Our first data source is for the crude rate from Ischaemic Heart Disease, the dependent variable Y. A crude rate is a measure of understanding the number of deaths in relation to the total population. We use the crude rate per 100,000 and calculate it for each state-year combination. It's calculated as the number of deaths from heart disease divided by the population of the state and multiplied by 100,000. A crude rate of 74 in Massachusetts in 2004 tells that if we took 100,00 people at random in Massachusetts in 2004, approximately 74 of them would die from Ischaemic Heart Disease.

In order to run our regression, we need data on the crude rate for all our states and years in the range 1999 to 2019. While this might ordinarily be a challenge because different states may classify deaths in different ways or some states may not have concrete records, the CDC stores exactly what we're looking for. The CDC Wonder Cause of Death database has the crude rate per 100,000 for all our states in our time period. Our initial concern with this data was that classifying the cause of death is difficult, and it can be hard to determine any single cause; however, the CDC has been using the 10th edition of the International Classification of Diseases (ICD) to classify death since 1999. Also, each death is only attributed to a single underlying condition, so we can be sure our data is reliable. As of this point, we have 1,071 rows of data, and each row has the crude rate for the corresponding state-year combination. This will serve as our skeleton to then add additional data, such as the unemployment rate for a given state in a given year.

Our second data source is for Medicaid Expansion, the independent variable X. As we are trying to find the effect of Expansion on mortality rate, Expansion is our treatment variable. For each state-year combination or row, we consider whether treatment has or has not happened. If treatment has happened, we assign a value of 1 and otherwise 0. For say Massachusetts in 2004, we would give treatment a value of 0 because Medicaid Expansion happened in 2014. Thus, for Massachusetts, the years 1999 to 2013 would have a treatment value of 0, and the years 2014 to 2019 would have a treatment value of 1. Importantly, another state does not necessarily have the same corresponding treatment values for those given years. Texas, for example, has not enacted Medicaid Expansion, so all Texas rows from 1999 to 2019 have a treatment value of 0.

Now that we understand what the treatment value is and how it's calculated for each row, we will examine how we extracted the data. In order to create our treatment column for all rows, we need the Medicaid Expansion year, or lack thereof, for all US states and DC. Fortunately, the Kaiser Family Foundation has a downloadable table of all states and DC and their expansion date. If a state didn't adopt Expansion, the cell says 'N/A.' From this table, we then extract the year from the date. The Massachusetts expansion year was 1/1/2014, so we simply extract the year 2014. Then, we go to our skeleton of 1071 rows, find Massachusetts, and add a 0 for all years before 2014 and 1 for the remaining years.

We are almost done with our data because we have statistics for our X and Y variables. The missing piece is that we also need data for our controls. This is because of omitted variable bias. If we were to run our regression on only Expansion and crude rate, we may overestimate the effect of Medicaid Expansion. As such, we instead want to run our regression while controlling for important factors such as the economy or specific policies. For this paper, we consider three controls.

First, we control for the unemployment rate with data from the US Bureau of Labor Statistics. The table provides the unemployment for all states and years in our time period. For Massachusetts in 2004, for example, the unemployment rate was 5.1%, so we add that to its corresponding row under the column for unemployment rate. We control for the unemployment rate because states with high unemployment may have fewer individuals with health insurance, and they may participate in more activities that could result in heart disease, such as smoking.

Second, we control for poverty with data from SAIPE (Small Area Income and Poverty Estimates), which is the Census Bureau's dataset for annual, state-level poverty estimates. This data set has the poverty rate for each state and year in our range. Unfortunately, each data set is only for 1 year, so we instead pull the data programmatically. Controlling for poverty is important because more impoverished states may benefit more from Expansion.

Finally, we control for policy by considering the cigarette tax for each state-year combination. The CDC has data on the State Tax per Pack from 1970 to 2019. Smoking cigarettes is highly correlated with heart disease, and higher taxes discourage people from smoking. Thus, by controlling for tax, we further limit the effect to Expansion and not taxes

We now examine our manipulations of the data considered. The first thing to consider is the time frame. We chose to start in 1999 because that's when the CDC Wonder portal began using the 10th edition of the ICD to classify causes of death. If we were to include data from 1998, causes of death may be identified differently for some diseases. We ended in 2019 to avoid the COVID years, as unfortunately, many deaths were a result of COVID and an underlying condition, such as heart disease. As it was an unprecedented time, we expect that the data is likely noisy. While we previously mentioned that we limited our age range to 55-64 because this

group is at risk of heart disease but doesn't yet qualify for Medicare, we ended up expanding to 45-64 because with a larger population to consider, our estimates became more precise.

Finally, we consider the full summary statistics from all our data sources for adults 45-64 from 1999 to 2019. This simply means looking at the mean, standard deviation, min, and max for our data points. For example, the highest crude rate was 180.49 in Mississippi in 1999, and the lowest was 42.43 in Utah in 2019. This gives two important insights. Firstly, from 1999 to 2019, the crude rate of Ischaemic Heart Disease has generally decreased. Secondly, the crude rate varies widely across states due to culture, health care institutions, and more. The full table can be found as Figure 1 in the section VIII. Appendix.

IV. Empirical Methods

The Empirical Methods section will explain our regression equation and what assumptions were made. We can perform a difference in difference regression due to the unique nature of our panel data, explained in the previous section. A regression equation is simply meant to estimate a value based on given data points. In our case, we hope to estimate the crude rate of Ischaemic Heart Disease for a US state in a specific year. Our given data points are the expansion year, unemployment rate, poverty rate, and state cigarette taxes. Thus, if we know a state adopted expansion in a prior year, has unemployment of 5%, a poverty rate of 10%, and state cigarette taxes of \$1, we can estimate what the crude heart might be. Below, highlighted in grey, is the regression equation.

$$cruderate = \beta_1 \cdot treat + \beta_2 \cdot unemp + \beta_3 \cdot pov + \beta_4 \cdot cig + u$$

Continuing our example, we would plug in 1 for “treat” because the state was treated with Expansion, .05 for “unemp” because of the 5% unemployment rate, and so on. Next are the

beta values, “ β_i ”. These beta values can be thought of as m, in the typical $y = mx + b$ slope format. For example, if “ β_1 ” were -20, it would imply that Expansion lowers the crude rate by 20 people. Finally, u is our error or difference from the true crude rate and our estimate.

We can improve upon the above regression by adding state and time fixed effects. We do this to account for differences between states and across time. For example, given Utah’s large population of Mormons who don’t smoke, it’s no surprise their crude rate is so low. Similarly, given technological advancements in health care, it’s no surprise that crude rates were significantly lower in 2019 than in 1999. While the above regression can not account for these distinctions, the regression below, with state and time fixed effects, can.

$$cruderate_{it} = \beta_1 \cdot treat_{it} + \beta_2 \cdot unemp_{it} + \beta_3 \cdot pov_{it} + \beta_4 \cdot cig_{it} + \beta_5 \cdot Z_i + \beta_6 \cdot S_t + u_{it}$$

This regression is similar to the previous one, but now has “it” subscripts and a few new variables. The “i” refers to a US state or DC, and the “t” refers to a year in our range. Thus, if we want the crude rate in Massachusetts in 2004, we’re looking for $cruderate_{Massachusetts, 2004}$. Similarly, we would plug in the corresponding treat, unemp, pov, and cig data for that state and year, and the beta values would act as before. The new parts are Z_i , the state fixed effect, and S_t the time fixed effect. Z_i is unique for each state just as S_t is unique to each year. Now, Utah would have a specific Z_i value that would better estimate the crude rate to account for its unique population. Finally, we cluster standard errors at the state level to allow for correlation within states because it’s expected that the same state will have similar crude rates to itself.

Now that we understand the regression equation, we consider the assumptions specific to this paper. First is the parallel trends assumption. Essentially, we want to show that before expansion, the crude rate of all states followed a similar trend. We first do this visually, then with

a canonical event study graph. For the visual check, we compare states that adopted Expansion in 2014 to those that never did. We'll consider the subset of men aged 55-64 as a representative sample because if there's a clear trend for this specific group, it'll also apply to the broader group of adults aged 45-64. The horizontal axis is the year, and the vertical axis is the crude rate. The full graph is under Figure 2 in Section VIII. Appendix.

There are three main takeaways from the line graph. Firstly, for both the 2014 Expansion states and the Never Expanded states, the crude rate generally decreases over the full time period. This suggests significant improvements in health outcomes over the period. That said, both rates start to flatten around 2011, meaning health advances seemed to hit a ceiling. Secondly, the 2014 Expansion states have a crude rate that is consistently, roughly 50 deaths lower than the non-expansion states. This fact is true before Expansion happens in 2014, so it implies that the 2 sets of states are systematically different in terms of culture and health care. This suggests plausible value in learning more about why non-Expansion states have worse health outcomes and what can be done to bring their numbers to their Expansion counterparts. Finally, and most importantly, we can't see a significant change in trend for either group after Expansion in 2014. While we'll still need to do an event study and formal regression, it suggests that we were not able to find a statistically significant effect.

For the formal event study, we consider all states and expand to all adults aged 45-64. Now, our X-axis is the years relative to Expansion. We consider 7 years before expansion, up to 5 years after Expansion. This means we can include states that adopted Expansion in different years and also include those that never did as controls. Additionally, instead of considering the exact crude rate, we consider the change in the crude rate. The vertical, dotted red line is right before Expansion happens. That's where we would expect to see a change in effect. The central

blue line is the average effect, and the highlighted blue area represents the 95% confidence interval. The full canonical event study can be found as Figure 3 in Section VIII. Appendix.

The key takeaway from the event study is that the effect remains around 0 after Expansion; however, the 95% confidence interval is relatively wide and expands from -5 to 5 before expansion to -5 to 10 after Expansion. For the identifying assumption of parallel trends, it is significant that the coefficients from before Expansion are all near 0. This is encouraging because it suggests that there were no diverging paths before Expansion. We do expect noisier data for years such as 5 years after expansion because only the subset of states that adopted Expansion in 2014 can be part of that data. Since our data ends in 2019, only the 50% of all states that adopted in 2014 can be used for the 5 years after expansion data in our event study. That said, however, the near-zero pre-treatment coefficients do support the parallel trends assumption because it implies the states followed similar trends before Expansion.

Thus, from the above line graph and canonical event study, we argue that we uphold parallel trends. This is the key identifying assumption for our difference in difference, and it argues that absent Expansion, treated and untreated states would have continued along similar trajectories for crude rate from Ischaemic Heart Disease. In addition, we uphold three additional identifying assumptions. First is Discrete Policy, which explains that treatment must happen at a specific time and not gradually. We believe we uphold this assumption because Expansion happens on a specific date, and our time frame is yearly, so there's enough time for people to access their health care. Second is minimal spillovers, which explains that treatment in one state should not affect another state. We similarly believe we uphold this assumption because Expansion in one state does not mean Expansion in neighboring states. Finally, we argue that

treatment is exogenous because no singular person had control over whether or not their state got Expansion.

V. Estimation

Now, we explore the results of our difference-in-difference regression. We'll share the results for our main group, two subgroups, and an additional interesting regression. By regressions, we simply mean estimating the best values of the beta, " β_i ", from before.

We begin with our main group of adults aged 45-64 and run a total of 4 regressions (full table in Figure 4 of VIII. Appendix). In our first regression, we control only for treatment. Essentially, we estimate the crude rate of a state in a given year based on if they adopted Medicaid Expansion. Note, in each subsequent regression, they each control for all variables mentioned previously. The second regression controls for treatment, as before, and the state and time fixed effects we explored earlier. While we don't report specific values for the state and time fixed effects, we know that having them makes the beta values more specific to each state. In the third regression, we add two economic controls. Specifically, we control for the unemployment rate and poverty rate for the reasons mentioned previously. In the final regression, we add the cigarette tax control.

To get into the values themselves, we'll begin with regression 1. For US adults aged 45-64, we find a coefficient on treatment of -14.51 that's statistically significant at the 5% level. This implies that states with Medicaid Expansion have 14.5 fewer deaths per 100,000 people from Ischaemic Heart Disease. It may initially seem that, because of this result, Expansion must lead to a substantial decrease in crude rate for this population; however, this result is due to the distinct circumstances of states that choose to adopt expansion and those that don't. More

specifically, states that adopted Expansion tend to have better healthcare systems and more education around health. Indeed, once we control for state and time fixed effects, in regression 2, the coefficient on treatment is near 0 and not statistically significant. The standard error of 2.3 is relatively large compared to the coefficient value of -0.4. Interpreting at face value implies that Expansion led to a half-person decrease in crude deaths per 100,000 people for heart disease while controlling for state and time fixed effects. Finally, note the large change in R-squared value between regression 1 and 2, which implies that state and time level characteristics explain much of the variation.

In regressions 3 and 4, while we begin to see coefficients that are significant at the 10% level, they're still too imprecise to draw precise conclusions. In regressions 3 and 4, the coefficient on unemployment is around -1 and significant at the 10% level. Holding all else constant, this implies a 1% increase in unemployment would lead to a 1 person decrease in crude rate. This is unexpected, as one might expect that adverse economic conditions would lead to worse health outcomes. That said, it's possible that a loss of a job may mean less money to participate in activities that could increase mortality, such as smoking. We see a similar trend for poverty rate, but the coefficient is not significant at even the 10% level. The cigarette tax coefficient did follow expectations, as a \$1 increase in taxes is associated with 1.3 fewer crude deaths. That said, the effect is not statistically significant due to the high standard error.

As a whole, holding state and time factors constant, Medicaid Expansion does not seem to be associated with a statistically significant change in crude rate from Ischaemic Heart Disease for adults aged 45 to 64. Importantly, this does not mean an effect does not exist. It simply suggests that an effect may be too small to see at the state level for the short time duration of 5 years after Expansion. Importantly, due to our 95% confidence interval, we can estimate a 95%

confidence interval for what the effect of Expansion was with all our controls. Given a coefficient of .508 and a standard error of 2.340, we can estimate that the effect of Expansion for adults 45 to 64 is between -4.1 and 5.1. Thus, while we can't specify the exact effect, we can rule out an effect of greater than 6 in either direction with 95% confidence.

While these regressions are important and do contribute to the literature, we explored additional regressions to see if any subgroup may offer more statistically significant results. In the next set of regressions, we limited our population to men only, and in the other set, we limited it to adults aged 55-65. This is because men develop heart disease about 7 years earlier than women and at a significantly higher rate (Journal of the American Heart Association 2026). The full regression can be found in Figure 5 in Section VIII. Appendix.

For the most part, the regression of men aged 45-64 is very similar to the regression for all adults, but there are a few key differences. Firstly, the coefficient on treatment for men in regression 1 is around -22, which is 50% higher than that for all adults. This suggests that the disparities between states are higher when considering the subgroup of men instead of adults as a whole. In light of this, the coefficient on unemployment is more negative and still statistically significant at the 10% level. Further suggesting that the relationship between unemployment and crude rate may be negative. Finally, the standard errors for treatment are larger, still near 0, and still statistically insignificant. This suggests our estimate is less precise for this group.

In this next set of regressions, we focus on adults aged 55-64 (Figure 6 in section VIII. Appendix). We expected this group to be more significantly affected by Expansion because they are more at risk of heart disease by their age. This data was consistent with this finding by the coefficient on treatment for regression 1. It had a value of -29, which is double the effect of -14.5 for adults in the wider age group of adults 45-64. This suggests further disparities among adults

aged 55 to 64 in Expansion vs. non-Expansion states. Unfortunately, the standard errors on treatment increased as well, meaning our results are less precise. Finally, the coefficients on unemployment, which were statistically significant at the 10% level in the previous 2 sets of regressions, were not statistically significant anymore. Although technically the number of observations is still 1071, the total number of deaths is smaller in magnitude when we limit the age range. As such, it's possible that there aren't enough data points for our regression to draw a clear conclusion.

The final regression we want to share is for all-cause mortality in adults aged 45-64 (Figure 7 in section VIII. Appendix). While we previously considered the crude rate from heart disease, this final regression is for the crude rate from any death. While such a regression may initially seem unrelated to our topic on chronic diseases, it highlights two key insights. Firstly, the treatment effect on regression 1 is near 0 and not statistically significant. This is surprising because in all previous regressions, the coefficient was significant at the one percent level and between a magnitude of 14 and 29. As such, it seemed there was a systematic difference between states that adopted Expansion and those that didn't. That said, for the subsequent regression, the coefficient on treatment is still not statistically significant.

Secondly, the coefficient on cigarette tax in regression 4 is -24 and statistically significant at the 1% level. A 1% significance suggests a very high correlation, and that a \$1 increase in tax would decrease crude deaths by 24 people. While we had expected the tax to be significant for heart disease, it seems it was most significant for all-cause mortality. This underscores the importance of state-level policies in promoting health.

VI. Conclusion

This paper estimated the effect of Medicaid Expansion on Ischaemic Heart Disease Mortality for adults aged 45 to 64 from 1999 to 2019. It used panel data to construct a difference-in-difference for multiple subgroups and multiple regressions with a varying number of controls. Overall, the paper failed to find a statistically significant effect that was distinguishable from zero. That said, by using confidence intervals, we were able to rule out an effect larger than 5 deaths per 100,000 people. This suggests that while we can't determine the specific effect of Expansion for this disease and subgroup, it likely was not groundbreaking in this case.

While this paper gave a big picture overview, its analysis may not have been specific enough to capture meaningful effects. As explained previously, only 20 million of the US's over 300 million people gained access to health insurance because of Medicaid Expansion. While these effects are real and significant for those affected, they may not be meaningful enough to see at the whole state level. Furthermore, Expansion may have had a smaller effect on older populations. More experienced adults may be more likely to have insurance than young adults because they may have better career benefits and may prioritize their health more. As such, a policy to give more people healthcare may not affect older adults as much. Finally, classifying cause of death is a complex process, and even if the guidelines haven't changed, different health professionals and states are bound to classify the same death in different ways.

Future research can build upon this work in several ways. Broadly, this could mean trying to find the effect for the subset of adults that gained healthcare through Expansion, instead of the entire state. It might also mean broadening the focus to health outcomes beyond mortality: earlier diagnosis, better adherence, or reduced financial hardship. Finally, given the highly significant coefficient on the cigarette tax, it might mean exploring the effect of other taxes.

VII. References

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VIII. Appendix

Figure 1: Summary Statistics: Heart Disease Mortality, Adults 45-64, 1999-2019

	(1)			
	Mean	SD	Min	Max
crude_rate	93.41	27.74	42.43	180.49
treat_post	0.17	0.38	0.00	1.00
unemployment_rate	5.43	1.98	2.10	13.50
poverty_rate	13.18	3.27	5.60	23.90
cigarette_tax_per_pack	1.14	0.92	0.03	4.94

Observations at the state-year level.

Equation 1: $crude_rate = \beta_1 \cdot treat + \beta_2 \cdot unemp + \beta_3 \cdot pov + \beta_4 \cdot cig + u$

Equation 2:

$$crude_rate_{it} = \beta_1 \cdot treat_{it} + \beta_2 \cdot unemp_{it} + \beta_3 \cdot pov_{it} + \beta_4 \cdot cig_{it} + \beta_5 \cdot Z_i + \beta_6 \cdot S_t + u_{it}$$

Figure 2:

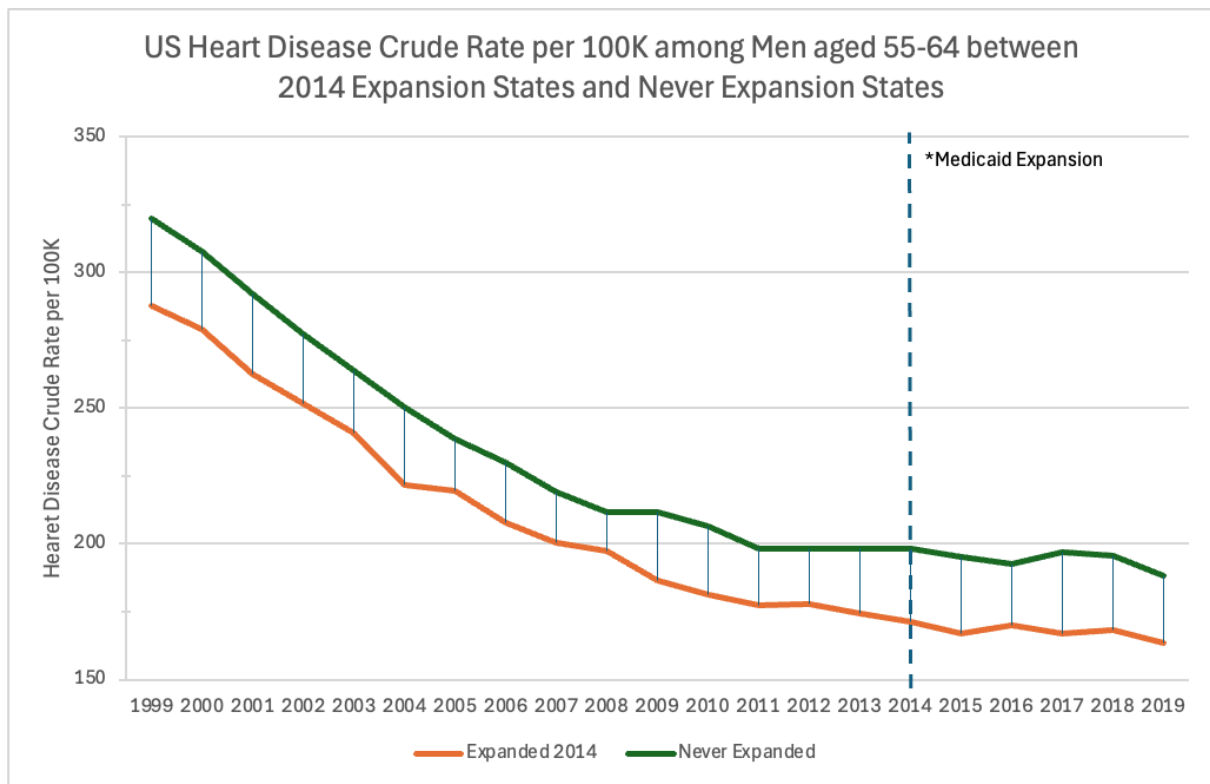


Figure 3:

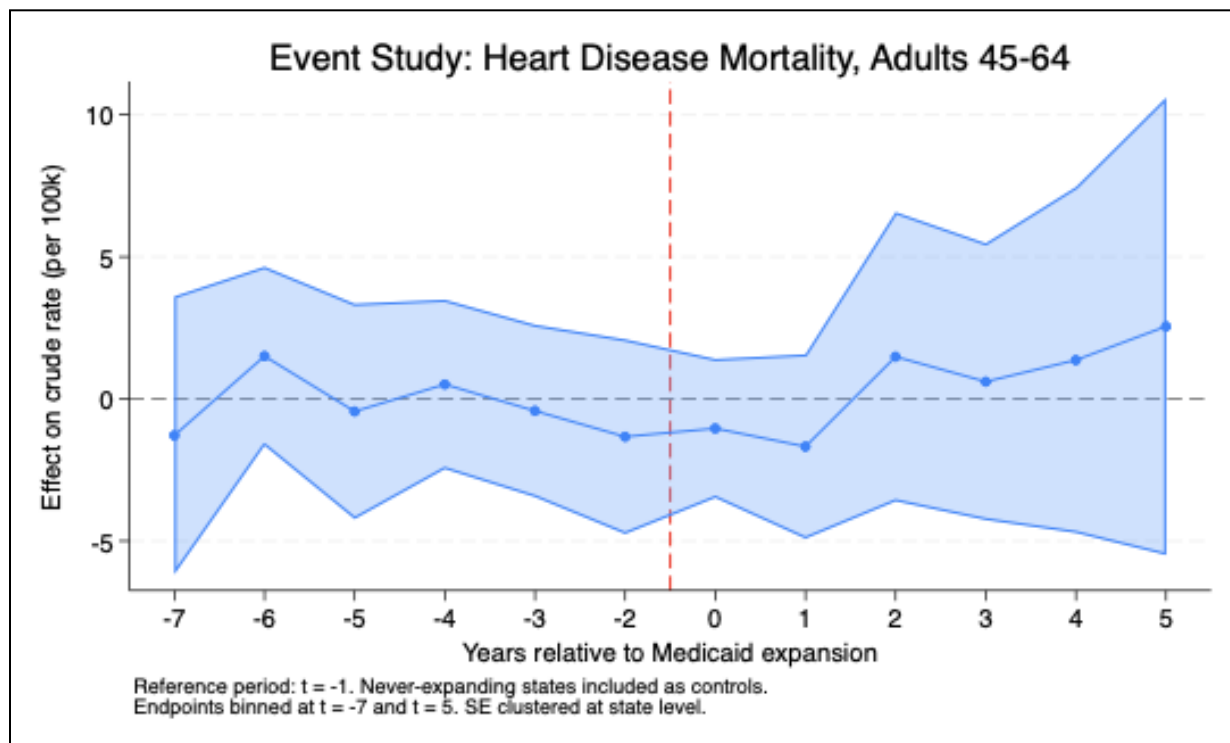


Figure 4:

DiD: Heart Disease Mortality, Adults 45-64

	(1)	(2)	(3)	(4)
	No FE	State+Year FE	+Econ Controls	+Cigarette Tax
treat_post	-14.51***	-0.408	-0.175	0.508
	(3.699)	(2.330)	(2.425)	(2.340)
unemployment_rate			-0.985*	-0.940*
			(0.549)	(0.525)
poverty_rate			-0.490	-0.564
			(0.768)	(0.761)
cigarette_tax_per_pack				-1.250
				(1.374)
State FE	No	Yes	Yes	Yes
Year FE	No	Yes	Yes	Yes

Observations	1071	1071	1071	1071
R-squared	0.039	0.927	0.928	0.929

Standard errors clustered at the state level.

Figure 5:

DiD: Heart Disease Mortality, Males 45-64

	(1)	(2)	(3)	(4)
	No FE	State+Year FE	+Econ Controls	+Cigarette Tax
treat_post	-21.97***	0.599	0.894	1.499
	(5.030)	(3.383)	(3.549)	(3.340)
unemployment_rate			-1.443*	-1.404*
			(0.821)	(0.793)
poverty_rate			-0.575	-0.640
			(1.177)	(1.164)
cigarette_tax_per_pack				-1.105
				(2.091)
State FE	No	Yes	Yes	Yes
Year FE	No	Yes	Yes	Yes
Observations	1071	1071	1071	1071
R-squared	0.047	0.909	0.910	0.910

Standard errors clustered at the state level.

Figure 6:

DiD: Heart Disease Mortality, Adults 55-64

	(1)	(2)	(3)	(4)
	No FE	State+Year FE	+Econ Controls	+Cigarette Tax
treat_post	-29.06***	-0.208	0.137	0.494
	(5.173)	(3.310)	(3.420)	(3.382)
unemployment_rate			-1.163	-1.139
			(0.743)	(0.721)

poverty_rate			-0.801	-0.839
			(1.008)	(1.030)
cigarette_tax_per_pack				-0.652
				(2.165)
State FE	No	Yes	Yes	Yes
Year FE	No	Yes	Yes	Yes
Observations	1071	1071	1071	1071
R-squared	0.064	0.925	0.926	0.926

Standard errors clustered at the state level.

Figure 7: DiD: All-Cause Mortality, Adults 45-64

	(1)	(2)	(3)	(4)
	No FE	State+Year FE	+Econ Controls	+Cigarette Tax
treat_post	1.544	-12.97	-14.45	-1.098
	(19.30)	(11.31)	(12.21)	(11.15)
unemployment_rate			-5.451*	-4.573*
			(2.719)	(2.451)
poverty_rate			5.956	4.516
			(5.213)	(4.217)
cigarette_tax_per_pack				-24.41***
				(6.831)
State FE	No	Yes	Yes	Yes
Year FE	No	Yes	Yes	Yes
Observations	1071	1071	1071	1071
R-squared	0.000	0.948	0.949	0.955

Standard errors clustered at the state level.

Figure 8: Medicaid Expansion Year for all US states and D.C.

Note: The year 3000 was used to give states that have not adopted expansion a numerical value.

State	Year	State	Year	State	Year
Alabama	3000	Maryland	2014	South Carolina	3000
Alaska	2015	Massachusetts	2014	South Dakota	2023
Arizona	2014	Michigan	2014	Tennessee	3000
Arkansas	2014	Minnesota	2014	Texas	3000
California	2014	Mississippi	3000	Utah	2020
Colorado	2014	Missouri	2021	Vermont	2014
Connecticut	2014	Montana	2016	Virginia	2019
Delaware	2014	Nebraska	2020	Washington	2014
District of Columbia	2014	Nevada	2014	West Virginia	2014
Florida	3000	New Hampshire	2014	Wisconsin	3000
Georgia	3000	New Jersey	2014	Wyoming	3000
Hawaii	2014	New Mexico	2014		
Idaho	2020	New York	2014		
Illinois	2014	North Carolina	2023		
Indiana	2015	North Dakota	2014		
Iowa	2014	Ohio	2014		
Kansas	3000	Oklahoma	2021		
Kentucky	2014	Oregon	2014		
Louisiana	2016	Pennsylvania	2015		
Maine	2019	Rhode Island	2014		